Oğuzhan Oğuz

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Profile

Master's student in Information and Network Engineering at KTH Royal Institute of Technology, following the Multimedia Processing and Analysis track, with a strong foundation in signal processing and communications from Middle East Technical University. Hands-on experience in Python and MATLAB across speech and audio processing, image and video processing, adaptive signal processing, pattern recognition, and machine learning. Interested in research and engineering roles that connect theory with practical multimedia, AI, and data-driven signal-processing systems.

Education

KTH Royal Institute of Technology — Aug 2025 – Present
M.Sc. in Information and Network Engineering — *Stockholm, Sweden*
Multimedia Processing and Analysis track. GPA: 5.0/5.0.

Middle East Technical University — Aug 2020 – Jun 2025
B.Sc. in Electrical and Electronics Engineering — *Ankara, Türkiye*
GPA: 3.57/4.00. Ranked 28/396 in class.

Experience

DEICO Engineering Inc. — Jan 2025 – Feb 2025
Test Engineering Intern — *Ankara, Türkiye*

- Developed C# interface applications for automated device control and testing, and implemented serial-communication workflows for reporting and validation.

National Magnetic Resonance Research Center — Jul 2024 – Aug 2024
Undergraduate Research Intern — *Ankara, Türkiye*

- Investigated jitter effects with mathematical modeling and MATLAB simulation under the supervision of Prof. Ergin Atalar, and documented results in a technical report.

Selected Projects

- **Inside-Out Tracking Sensor Suite** *(B.Sc. Graduation Project)*: Developed a real-time inside-out VR tracking pipeline using camera-IMU fusion for 6-DoF head pose estimation; implemented low-latency Wi-Fi/UDP communication and Python-based external rendering, and contributed to the image-processing pipeline.
- **Signal Classification with Deep Learning**: Implemented a 1D residual neural network to classify five analog modulation types from synthetic I/Q data under varying SNR and carrier-offset conditions.
- **Video Coding with Motion Compensation**: Built and compared intra, conditional replenishment, and motion-compensated codecs; evaluated bitrate-PSNR trade-offs on QCIF video sequences.
- **Wiener Filtering for Speech Enhancement**: Designed and compared non-causal, FIR, and causal Wiener filters using AR-based speech and noise modeling from a single noisy recording.

Technical Skills

Programming: Python, MATLAB, C#
Libraries and Frameworks: PyTorch, NumPy, SciPy, Matplotlib, OpenCV
Technical Areas: Data Science, Signal Processing, Adaptive Signal Processing, Speech and Audio Processing, Image and Video Processing, Multimedia Processing, Pattern Recognition, Machine Learning, Digital Communications
Tools and Platforms: Git, Linux, LaTeX, Siemens NX

Languages

Turkish: Native — **English:** C1 listening/reading, B2 speaking/writing (IELTS 7.0)

Intended position: Research internship, research assistantship, or entry-level role in data science, signal processing, image/video processing, speech/audio processing, or multimedia processing. **Country:** Sweden.